

Task A: BCNF Normalization

Food Pantry Database for BCNF Compliance

The relational schema of the food pantry database has been meticulously designed to adhere to the principles of Boyce-Codd Normal Form (BCNF), a stringent form of normalization that ensures the elimination of redundancy, update anomalies, and inconsistencies. BCNF is characterized by the requirement that for every non-trivial functional dependency ($A \rightarrow B$), (A) must be a superkey. Below is an analysis of each table within the database to affirm its compliance with BCNF:

1. Admin Table

- **Structure:** Comprises Admin_Id, Employee_Name, Employee_Email_Address, Employee_Phone_Number, and Password.
- **Primary Key:** Admin_Id.
- **Functional Dependency:** Admin_Id functionally determines all other attributes in the table.
- **BCNF Compliance:** Given that Admin_Id is the primary key and the sole determinant, all attributes are fully functionally dependent on it. Hence, the Admin table satisfies the conditions of BCNF.

2. Students Table

- **Structure:** Consists of Student_Id, Student_Name, Student_Email_Address, Student_Phone_Number, and Password.
- **Primary Key:** Student_Id.
- **Functional Dependency:** Student_Id uniquely determines all other attributes.
- **BCNF Compliance:** As Student_Id is the primary key, and there are no partial or transitive dependencies, the Students table adheres to the BCNF standards.

3. Products Table

- **Structure:** Includes Product_Id, Product_Name, Product_Category_Id, Product_Quantity, and image_path.
- **Primary Key:** Product_Id.
- **Functional Dependency:** Product_Id is the determinant for all other attributes.
- **BCNF Compliance:** The Products table is in BCNF as all attributes are fully functionally dependent on the primary key, Product_Id, with no extraneous dependencies.

4. Categories Table

- **Structure:** Contains Category_Id, Category_Name, and Category_Description.
- **Primary Key:** Category_Id.
- **Functional Dependency:** Category_Id functionally determines both Category_Name and Category_Description.
- **BCNF Compliance:** The Categories table meets BCNF criteria as the primary key, Category_Id, is the only determinant, ensuring full functional dependency.

5. Cart Table

- **Structure:** Comprises Order_Id, Product_Id, Student_Id, Products_Quantity, Admin_Id, and Is_Pickedup.
- **Primary Key:** Order_Id.
- **Functional Dependency:** Order_Id determines all other attributes in the table.
- **BCNF Compliance:** The Cart table aligns with BCNF principles since Order_Id, as the primary key, is the sole determinant, and all attributes are fully dependent on it.

In conclusion, each table within the food pantry database is structured to comply with the BCNF. This adherence ensures the integrity and efficiency of the database, minimizing redundancy and potential anomalies in data management. The design reflects a robust and well-thought-out approach to database normalization, crucial for maintaining the quality and reliability of the database system.